Lecture 4

Thursday, 31 August 2023 10:34 AM

Chapter 8 ; Niemantsverdriet ; Bulk vs. Surface-sensitive ;

DRIFTS · Diffuse Reflectance IR (DRIFTS)

+ Transmission needs the IR to pass through (self-supporting wafer required to be input in the sample)

+ But when support or catalyst is very dark & opaque → DRIFTS
  ie. powder properties
  ↓
  More towards the surface technique

+ Challenges : In-situ vs. operando
  ↓ reaction conditions
  ↳ Exactly how its done

+ Sample is "infinitely thick"
  ↳ But cannot be too thick since there will be significant gradients

Drifts issues : T gradients, diffusion resistance (Conc. gradient due to lack of porosity)

Drifts is not as quantitative as Transmission

+ Attenuated Total Reflectance (ATR) -IR:
  + Planar model catalysts, thin films
  + Water absorbs a lot of IR
  + Can be used for liquid-solids interfaces

+ UV-vis and Raman Spectroscopy

↑ wave → ↑ energy → ↓ wavelength

LMCT → Ligand-to-metal-charge-transfer

MLCT → metal-to-ligand "

d–d transitions : lower energy

CT → Higher

ε → Absorbsivities (Signal value)

Spectra → can you explain all the peaks & what peaks are not there.

Raman Spectroscopy (RS)

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Inelastic scattering; when light is incident

IR only gives single level transitions → Using FT to figure out which one got adsorbed

RS

+ choose a specific wavelength

- Rayleigh scattering
- Stokes shift
- Anti-stokes

Vibrational transitions not the same as electronic transitions

↓

Raman Shifts: you can figure out identity of material

UV Raman

Removes flourescence of adsorbed Carbon species